

Subject: CSE - 427, Digital Image Processing

Time: 3.00 hours

Full Marks:180

INSTRUCTIONS:

- a. Use SEPARATE answer scripts for each section.
- b. Question-1 in section A and Question-5 in section B are compulsory.
- c. Answer any OTHER TWO questions from EACH section.
- d. Figures in the margin indicate full marks.
- e. Assume reasonable data if necessary.
- f. Symbols used have their usual meanings.

SECTION-A

Question – 1

- a. Explain the following terminologies: 06
 - (i) Radiance
 - (ii) Luminance
 - (iii) Brightness
- b. Justify as to why ultraviolet rays or soft X-rays are required to study water molecules. 04
- c. Suppose you want to create an image of size [1024X512] having 72 distinct intensity values. What will be the size of the image. 04
- d. Differentiate between image enhancement in the spatial domain and frequency domain. 04
- e. Let s, r are all discrete and in the range $[0, L-1]$. Determine the transformation function for Histogram Equalization and discuss the two conditions which the transformation function must adhere to. 04
- f. Find out the issues in spatial filtering when the mask moves closer to the border. Also, indentify the way outs to resolve the discovered issue. 04
- g. Interpret the properties of the 1st order and 2nd order derivatives with the example of an image strip. 04

Question – 2

- a. You are hired to design the front end of an imaging system for studying the boundary shapes of cells, bacteria, viruses and protein. The front end consists in this case, of the illumination source and corresponding imaging camera. The diameters of circles required to enclose individual specimens in each of these categories are 25, 0.5, 0.05 and 0.005 μm respectively. 06

Can you solve the imaging aspects of this problem with a single sensor and camera? If your answer is 'yes', specify the illumination wavelength band and type of camera needed.

By type, the band of the electromagnetic spectrum to which the camera is most sensitive is meant (e.g., infrared).

If your answer is 'no', what type of illumination source and corresponding imaging sensors would you recommend?
- b. Consider the two image subsets S_1 and S_2 shown in the figure 01. For $V = \{1\}$, determine whether these two image subsets are 04
 - (i) 4-adjacent
 - (ii) 8-adjacent
 - (iii) m-adjacent

Figure 01 to Q-2(b)

	S_1				S_2				
0	0	0	0	0	0	0	1	1	0
1	0	0	1	0	0	1	0	0	1
1	0	0	1	0	1	1	0	0	0
0	0	1	1	1	0	0	0	0	0
0	0	1	1	1	0	0	1	1	1

- c. An automobile manufacturer is automating the placement of certain components on the bumpers of a limited-edition line of sports cars. The components are color coordinated, so, the robots need to know the color of each car in order to select the appropriate bumper component. Models come only in four colors – blue, green, red and white. You are hired to propose a solution based on imaging. How would you solve the problem of automatically determining the color of each car, keeping in mind the cost is the most important consideration in your choice of components? 10
- d. Consider two 8-bit images whose intensity levels span the full range from 0 to 255. 10
- (i) Discuss the limiting effect of repeatedly subtracting the image (2) from (1). Assume that the result is represented also in eight bits.
- (ii) Would reversing the order of images yield a different result?

Question – 3

- a. Propose a set of intensity-slicing transformations capable of producing all the individual bit planes of an 4-bit monochrome image. (For example, a transformation function with the property $T(r) = 0$ for r in the range $[0,7]$ and $T(r) = 15$ for r in the range $[8, 15]$ produces an image of the 4th bit plane in an 8 bit image.) 10
- b. Justify why the discrete histogram equalization technique does not, in general, yield a flat histogram. 05
- c. Suppose that the digital image is subjected to histogram equalization. Show that a second pass of histogram equalization (on the histogram equalized image) will produce exactly the same result as the first pass. 08
- d. In a given application, an averaging mask is applied to insert images to reduce noise, and then a laplacian mask is applied to enhance the small details. Would the result be the same if the order of these operations were reversed? 07

Question – 4

- a. A Laplacian mask with a (-8) in the center yields sharper results than the one with a (-4) in the center. Examine the reason in detail by taking example of a sample image and suitable laplacian mask. 10
- b. Suppose that you filter an image, $f(x,y)$ with a spatial mask filter, $w(x,y)$, using convolution, where the mask is smaller than the image in both spatial directions 10
- (i) Show that, if the coefficients of the mask sum to zero, then sum of all the elements in the resulting convolution array will be zero too. You may assume that the border of the image has been padded with appropriate number of zeros.
- (ii) Would the result to (i) be the same if filtering is implemented using correlation?
- c. (i) Give the conditions under which the D_4 distance between two points p & q is equal to the shortest 4-path between these two points. 10
- (ii) Is this path unique?

SECTION-B

Question - 5

- a. Consider the [4X8] image of 8 bit. 15
- | | | | | | | | |
|----|----|----|----|-----|-----|-----|-----|
| 21 | 21 | 95 | 95 | 169 | 169 | 243 | 243 |
| 21 | 21 | 95 | 95 | 169 | 169 | 243 | 243 |
| 21 | 21 | 95 | 95 | 169 | 169 | 243 | 243 |
| 21 | 21 | 95 | 95 | 169 | 169 | 243 | 243 |
- (i) Compute the entropy of the image.
(ii) Compress the image using Huffman Coding.
(iii) Compute the compression achieved and the effectiveness of Huffman Coding.
- b. What is 2D discrete fourier transform? Discuss the need of transforming an image into its frequency domain. 08
- c. Small changes in intensity can have significant impact on the complexity of the bit planes. Justify the above statement with an example. Also, bring out the solution to the above problem. 07

Question -6

- a. Discuss the Ideal low pass filter and Ideal high pass filter with suitable examples and figures. 10
- b. Discuss in detail about the steps in image filtering in the frequency domain. 05
- c. What are the properties of fourier transform? List equations if any. 05
- d. Discuss the image compression process in detail with respect to the following 10
- (i) Mapper
 - (ii) Quantizer
 - (iii) Symbol coder
 - (iv) Inverse Mapper
 - (v) Decoder

Question -7

- a. The arithmetic decoding procedure is the reverse of the encoding procedure. Decode the message 0.32256, given the coding model below: 15

Symbol	Probability
a	0.2
e	0.3
i	0.1
o	0.2
u	0.1
!	0.1

- b. Use the LZW coding algorithm to encode the 7-bit ASCII string below: 15
- "AAAAAAAAAAAA"

Question-8

- a. A binary image contains straight lines oriented horizontally, vertically at 45° and -45°. 15
- Give a set of 5X5 masks that can be used to detect 1 pixel breaks in these lines. Assume that the intensities of the lines and background are 1 & 0, respectively.
- b. In an image automated scoring system project of a 10 m pistol firing image, you need to find out the score of a firer. The target is 15

depicted below in figure 8(b).

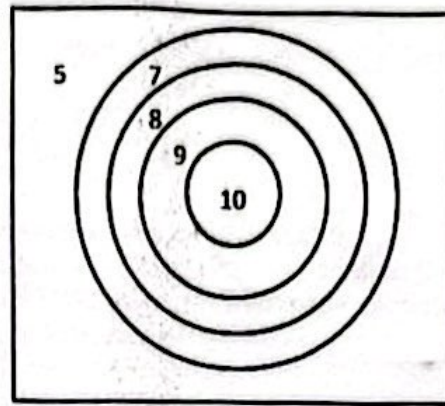


Figure 8(b)

When a bullet hits the target, a hole is formed in the board. A camera is placed beneath the target which clicks the image of the target after firing of 10 rounds is completed.

You need to process the captured image and find out the score of the firer out of 100. Now answer the following questions:

- (i) Illustrate the concepts of image processing to be used to capture the bullet hits in the target. Design specific masks to substantiate your justification.
- (ii) After the hits have been identified, you need to find the total score. Which concept of image processing will be used in this case

BANGLADESH UNIVERSITY OF PROFESSIONALS

Military Institute of Science and Technology

B.Sc. In Computer Science and Engineering

Short Term Final Examination: Jan 2019

Subject: CSE-427, Digital Image Processing

Time: 3.00 hours

Full Marks: 210

INSTRUCTIONS:

- Use **SEPARATE** answer scripts for each section.
- Answer any **THREE** questions from each section out of **FOUR**.
- Figures in the margin indicate full marks.
- Assume reasonable data if necessary.
- Symbols** used have their usual meanings.

SECTION-A

Question - 1

- a. Define the following terms :-

5X4=20

- Digital Image
- Luminance
- Intensity Resolution
- Spatial Resolution
- False Contouring

- b. With necessary examples show that imaginary is possible in any range of the entire EM spectrum though human beings can see only in a narrow portion of EM spectrum.

10

- c. When an operation is called linear? Show that the maximum operation is not linear.

5

Question - 2

- a. The following are the gray scale values of a scan line of an image. EOB is a special symbol representing the end of the image block. Compress the segment using

25

- Huffman Coding [Ignore EOB symbol]
- Arithmetic Coding

Is arithmetic coding lossless? Justify.

217 17 223 223 223 18 EOB

- b. Do you really need to implement algorithms for 2D Fourier transform and its inverse to manage the spectrum of an image? Justify.

10

Question - 3

- a. What is the blocking effect in image compression? Write down the relative advantages and disadvantages of KL (Karhunen-Loeve) transform, DFT (Discrete Fourier Transform), DCT (Discrete Cosine Transform) and Walsh-Hadamard Transformation (WHT) in image compression.

15

- b. What is contrast enhancement of images? Explain one of its uses in medical imaging.

10

- c. In order to generate digital images from secured data, sampling and quantification are the important processes. How do these operations affect the size and color information of an image?

10

Question - 4

- a. Explain the imaging procedure that is needed in ultrasonography (USG)

10

- b. Give the mathematical equation representing the convolution of a filter $W(X,Y)$ with an image $f(X,Y)$. Show the results of applying a

15

Box filter on an image of size 5X5 pixels with gray scale intensity values. Explain some of the convolution responses with that filter.

- c. How was the sobel mask designed for computing the gradient of an image?

SECTION-B

Question -5

- a. During image enhancement using image subtraction method, we can end up having negative value for a pixel. Propose two solutions to overcome it.
- b. Derive the expression for first order and second order derivative at X in the horizontal intensity profile of a 2D image. Calculate the first order and second order derivatives at each point of the following 2-d image strip. By analyzing these values, identify ramp, line and isolated noise with explanation.

5	5	4	3	2	1	0	0	0	6	0	0	0	0	1	3	1	0	0	0	7	7	7	7
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---
- c. Which type of filter is most suitable to remove salt and pepper noise? What are the advantages of this filter?

Question - 6

- a. What is a gradient image? Provide a computationally suitable approximation of gradient image. Write down sobel gradient operations.
- b. Write down the steps for edge linking using local processing approach.
- c. Explain the Hough transform approach for edge linking. How is an alternative representation of lines used to mitigate the problem caused by infinite slope of lines?
How are the accumulators cells initialized and updated?

Question - 7

- a. Write down an algorithm for finding polygonal fitting of open and closed curves.
- b. Describe the region splitting and merging technologies for image segmentation. Which type of data structure is need for this purpose?
- c. What is an accumulative difference image (ADI)? How can this be needed to detect motion from an image?
- d. Write down the generalized global thresholding equation. What are the limitations of basic global thresholding?

Question - 8

- a. Write short notes on the following.
 - (i) HIS color model
 - (ii) CMYK color model

- (i) Gaussian High Pass filter (GHPF)
- (ii) Butterworth Low Pass Filter (BLPF)

c. Name different image sensor arrangements. Describe the one that is used in CT/PRT. Justify whether the resolution in this arrangement can be adjusted.

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BANGLADESH UNIVERSITY OF PROFESSIONALS
Military Institute of Science and Technology
B.Sc. in Computer Science and Engineering, Term Final (Spring) Examination 2021: July-Aug 2021

Student Group: Earned Credit Hours ≥ 108

Subject: CSE-427, Digital Image Processing

Total	: 3.00 hours (For Physical Examination)	Full Marks	: 180
	: 2.00 hours (Online Examination for Section-A + Section-B)		
Section B	: 1.00 hour (Online Examination)	Section B	: 90

INSTRUCTIONS - PHYSICAL EXAMINATION

- a. Use SEPARATE answer scripts for each section.
- b. Question - 5 in Section-B is compulsory.
- c. Answer any OTHER TWO questions from Section-B.
- d. Figures in the margin indicate full marks.
- e. Assume reasonable data if necessary.
- f. Symbols and abbreviations used have their usual meanings.

INSTRUCTIONS - ONLINE EXAMINATION (IF PHYSICAL EXAMINATION IS NOT HELD)

- a. Question - 5 and Question - 8 OR (Viva Voce) in Section B are compulsory.
- b. Answer any OTHER ONE question from Q-6 & Q-7 of Section-B.
- c. Instructions mentioned in d, e and f for physical examination are also applicable for Online Examination.

SECTION-B

Question - 5 (Compulsory)

- a. What is/are the basic difference/differences between the selection of conventional threshold value with optimal threshold value? 08
- b. Assume you have an image block and a structuring element shown below. Perform morphological closing operation and show the result for each step. 22

0	0	0	0	0	0	0	0
0	1	0	0	0	0	1	0
0	0	1	0	0	1	0	0
0	0	0	1	1	0	0	0
0	0	0	1	1	0	0	0
0	0	1	0	0	1	0	0
0	1	0	0	0	0	1	0
0	0	0	0	0	0	0	0

Image block

1	1	1	1
1	1	1	1
1	1	1	1

Structuring element

Question - 6

- a. Assume that you are given a RGB color image for processing. Pre-processing outcome of this given RGB color image using RGB color model and another color model (i.e., HSV) will produce same outcome after enhancement operation. Give your reason/reasons if you agree or disagree with the statement. 10
- b. Selection of an optimal threshold value is essential for Binary segmentation. Give your reason/reasons if you agree or disagree with the statement. 12
- c. What may happen if over-segmentation and/or under-segmentation occurs after segmentation? Explain with necessary figure/figures. 08

Question - 7

- a. How can you process a color image while converting it a gray-scale image? 10

- b. Perform x-correlation and x-convolution for the following.

10	20	10
40	50	50
10	20	10

Test image

10	5	10
2.5	2	2
10	5	10

Structuring element

Question – 8

- Define segmentation. Why do we need segmentation for conventional image processing?
- Suppose you have a RGB color Image. You need to perform double threshold operation for the higher part of the RGB image. Explain in details how you can do that.
- Use of pseudo-color is somewhat essential for digital image processing. Give your reason/reasons if you agree or disagree with the statement.

OR

VIVA VOCE (Compulsory in case of Online Examination)

Subject: CSE-427, Digital Image Processing

Total: 2.00 hours
Section A : 1.00 hour

Full Marks: 180
Section A : 90

INSTRUCTIONS:

- Use SEPARATE answer scripts for each section.
- Question – 3 and Question – 4 (Viva Voce) in Section A are compulsory.
- Answer any OTHER ONE question from this section (From Q - 1 & Q - 2).
- Figures in the margin indicate full marks.
- Assume reasonable data if necessary.
- Symbols used have their usual meanings.

SECTION-A

Question – 1

- You have an 8 bit gray scale image. What will happen for your 8 bit image histogram if you set the lower order bit planes to zero on the histogram? **08**
- You have an 8 bit gray scale image. What will happen for your image histogram (8 bit) if you set the higher order bit planes to zero on the histogram? **08**
- Define histogram equalization. Typically, it has been observed that discrete histogram equalization technique produces a non-smooth histogram. Justify your answer. **14**

Question – 2

- Calculate and show the resultant values of each pixel from the below chart after the first pass of histogram equalization approach. What will happen if you apply second pass histogram equalization approach? **20**

52	55	61	154
63	195	90	126
62	55	113	204
61	63	68	126

- What will happen if you continually apply a low-pass Spatial filter mask (5 x 5) while ignoring the border effects to a digital image? Do you think it will remain same if you apply 3 x 3 mask instead of 5 x 5 or it will change? **05**
- You have a digital image and two different masks (Average mask and Laplacian mask). For the given image, firstly you have implemented average mask to reduce noise and finally implemented Laplacian mask to reduce small details. What will happen if these masks are reversed for your implementation for the same image? **05**

Question – 3 (Compulsory)

Consider the following image strip.

5 5 4 3 2 1 0 0 0 6 0 0 0 0 1 3 1 0 0 0 0 7 7 7 7

- Evaluate first and second order derivatives from the above image **20**

strip. Point out gray level, step and ramp for first and second order derivatives.

b. Estimate with justification the following from the above strip (if any available)

- i. Gray level profile
- ii. Ramp
- iii. Step
- iv. Thin line Segment
- v. Isolated Point

Question - 4 (Compulsory)
Marks allotted for Viva 30

Subject: CSE-427, Digital Image Processing

Total: 2.00 hours
Section B : 1.00 hour

Full Marks: 180
Section B : 90

INSTRUCTIONS:

- Use SEPARATE answer scripts for each section.
- Question – 5 and Question – 8 (Viva Voce) in Section B are compulsory.
- Answer any OTHER ONE question from this section (From Q - 6 & Q - 7).
- Figures in the margin indicate full marks.
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- Symbols used have their usual meanings.

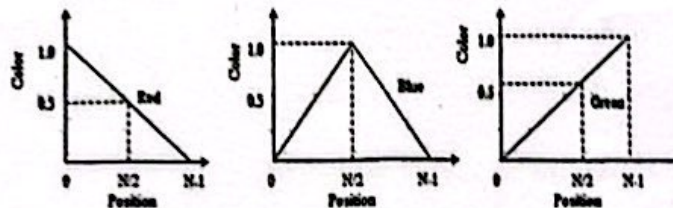
SECTION-B

Question – 5 (Compulsory)

- Define Spatial Correlation Convolution. 05
- Solve the following using Spatial Correlation and Convolution. 18

$\nwarrow$ Origin f 0001110101	w 12328
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- In a simple RGB image, the R, G and B component images have the horizontal intensity profiles shown in the below diagram. What color would a person will see in the middle column of this image? 07



Question – 6

- Define structuring element. Explain with example / examples about "Fits", "hits" and "neither Fits nor hits" associated with morphological image processing. 10
- An image with a structuring element is given below. What will be the result if you apply morphological closing operation in the given image? 20

0	0	0	0	0	0	0	0	0
0	0	1	1	0	1	1	0	0
0	1	1	1	1	1	1	1	0
0	0	1	1	0	0	1	1	0
0	1	0	0	0	1	1	1	0
0	1	1	0	1	0	1	1	0
0	1	0	1	1	1	0	0	0
0	0	0	0	0	0	0	0	0

Image

	1	
1	1	1
	1	

Structuring Element

Question – 7

a. Describe the followings:

- i. Hue
- ii. Pseudo-color Processing
- iii. Intensity
- iv. Tint
- v. Shade

b. Define segmentation. Describe with necessary figure how you can calculate multilevel threshold value for binary segmentation.

Question – 8 (Compulsory)

Marks allotted for Viva 30

BANGLADESH UNIVERSITY OF PROFESSIONALS
Military Institute of Science and Technology
B.Sc. in Computer Science and Engineering, Term Final (Spring) Examination 2021: July-Aug 2021

Student Group: Earned Credit Hours $\geq$ 108

Subject: CSE-427, Digital Image Processing

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	: 2.00 hours (Online Examination for Section-A + Section-B)		
Section B	: 1.00 hour (Online Examination)	Section B	: 90

INSTRUCTIONS - PHYSICAL EXAMINATION

- Use SEPARATE answer scripts for each section.
- Question - 5 in Section-B is compulsory.
- Answer any OTHER TWO questions from Section-B.
- Figures in the margin indicate full marks.
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INSTRUCTIONS - ONLINE EXAMINATION (IF PHYSICAL EXAMINATION IS NOT HELD)

- Question - 5 and Question - 8 OR (Viva Voce) in Section B are compulsory.
- Answer any OTHER ONE question from Q-6 & Q-7 of Section-B.
- Instructions mentioned in d, e and f for physical examination are also applicable for Online Examination.

SECTION-B

Question - 5 (Compulsory)

- What is/are the basic difference/differences between the selection of conventional threshold value with optimal threshold value? 08
- Assume you have an image block and a structuring element shown below. 22
Perform morphological closing operation and show the result for each step.

0	0	0	0	0	0	0	0
0	1	0	0	0	0	1	0
0	0	1	0	0	1	0	0
0	0	0	1	1	0	0	0
0	0	0	1	1	0	0	0
0	0	1	0	0	1	0	0
0	1	0	0	0	0	1	0
0	0	0	0	0	0	0	0

Image block

1	1	1	1
1	1	1	1
1	1	1	1

Structuring element

Question - 6

- Assume that you are given a RGB color image for processing. Pre-processing outcome of this given RGB color image using RGB color model and another color model (i.e., HSV) will produce same outcome after enhancement operation. Give your reason/reasons if you agree or disagree with the statement. 10
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- What may happen if over-segmentation and/or under-segmentation occurs after segmentation? Explain with necessary figure/figures. 08

Question - 7

- How can you process a color image while converting it a gray-scale image? 10

b. Perform x-correlation and x-convolution for the following.

10	20	10
40	50	50
10	20	10

Test image

10	5	10
2.5	2	2
10	5	10

Structuring element

Question – 8

- Define segmentation. Why do we need segmentation for conventional image processing?
- Suppose you have a RGB color Image. You need to perform double threshold operation for the higher part of the RGB image. Explain in details how you can do that.
- Use of pseudo-color is somewhat essential for digital image processing. Give your reason/reasons if you agree or disagree with the statement.

OR

VIVA VOCE (Compulsory in case of Online Examination)

Subject: CSE-427, Digital Image Processing

Total: 2.00 hours
Section A : 1.00 hour

Full Marks: 180
Section A : 90

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SECTION-A

Question – 1

- You have an 8 bit gray scale image. What will happen for your 8 bit image histogram if you set the lower order bit planes to zero on the histogram? **08**
- You have an 8 bit gray scale image. What will happen for your image histogram (8 bit) if you set the higher order bit planes to zero on the histogram? **08**
- Define histogram equalization. Typically, it has been observed that discrete histogram equalization technique produces a non-smooth histogram. Justify your answer. **14**

Question – 2

- Calculate and show the resultant values of each pixel from the below chart after the first pass of histogram equalization approach. What will happen if you apply second pass histogram equalization approach? **20**

52	55	61	154
63	195	90	126
62	55	113	204
61	63	68	126

- What will happen if you continually apply a low-pass Spatial filter mask (5 x 5) while ignoring the border effects to a digital image? Do you think it will remain same if you apply 3 x 3 mask instead of 5 x 5 or it will change? **05**
- You have a digital image and two different masks (Average mask and Laplacian mask). For the given image, firstly you have implemented average mask to reduce noise and finally implemented Laplacian mask to reduce small details. What will happen if these masks are reversed for your implementation for the same image? **05**

Question – 3 (Compulsory)

Consider the following image strip.

5543210006000013100007777

- Evaluate first and second order derivatives from the above image **20**

strip. Point out gray level, step and ramp for first and second order derivatives.

- b. Estimate with justification the following from the above strip (if any available)
- i. Gray level profile
 - ii. Ramp
 - iii. Step
 - iv. Thin line Segment
 - v. Isolated Point

Question – 4 (Compulsory)
Marks allotted for Viva 30